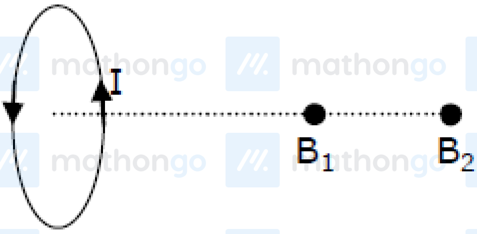
Q1 (3)

When a rod is free and it is heated then there is no thermal stress produced in it. The rod will expand due to increase in temperature. so both a \& R are true.

Q2 (3)



$B=\frac{\mu_0 N i R^2}{2\left(R^2+x^2\right)^{3/2}}$

at $\mathrm{x}_1=0.05 \mathrm{~m}, \mathrm{~B}_1=\frac{\mu_0 \mathrm{NiR}^2}{2\left(\mathrm{R}^2+(0.05)^2\right)^{3/2}}$

at $\mathrm{x}_2=0.2 \mathrm{~m}, \mathrm{~B}_2=\frac{\mu_0 \mathrm{NiR}^2}{2\left(\mathrm{R}^2+(0.2)^2\right)^{3/2}}$

$\frac{B_1}{B_2}=\frac{\left(R^2+0.04\right)^{3/2}}{\left(R^2+0.0025\right)^{3/2}}$

$\left(\frac{8}{1}\right)^{2/3}=\frac{R^2+0.04}{R^2+0.0025}$

$4\left(R^2+0.0025\right)=R^2+0.04$

$3R^2=0.04-0.0100$

$\mathrm{R}^2=\frac{0.03}{3}=0.01$

$\mathrm{R}=\sqrt{0.01}=0.1 \mathrm{~m}$

Q1: 25 Feb (Shift 1) - Single Correct

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : When a rod lying freely is heated, no thermal stress is developed in it. Reason R : On heating, the length of the rod increases. In the light of the above statements, choose the corect answer from the options given below :

(1) A is true but R is false

(2) Both A and R are true and R is the correct explanation of A

(3) Both A and R are true but R is NOT the correct explanation of A

(4) A is false but R is true

Q2: 25 Feb (Shift 1) - Single Correct

Magnetic fields at two points on the axis of a circular coil at a distance of 0.05 m and 02 m from the centre are in the rato 8: 1. The radius of coil is

(1) 0.15 m

(2) 0.2 m

(3) 0.1 m

(4) 1.0 m

Answer Key

Q1 (3) **Q2** (3)